

PAMPLONA AP, Spain

WMO: 080850

Lat: 42.7770N

Lon: 1.6498W

Elev: 1504

StdP: 13.91

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	28.3	17.7	14.2	32.7	21.3	16.8	32.7	25.8	39.2	23.2	40.9	5.3	340	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.1	94.8	67.5	91.1	67.1	87.2	66.3	70.6	86.1	69.1	84.4	67.7	82.6	8.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	101.2	75.3	64.4	95.6	74.3	62.8	90.5	73.2	35.4	85.7	34.0	84.4	32.9	82.6	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.8	20.4	18.4	DB	21.5	100.5	3.5	2.7	19.0	102.5	16.9	104.1	15.0	105.6	12.4	107.6	(4)
			WB	20.1	73.8	3.6	1.6	17.6	74.9	15.5	75.9	13.5	76.8	10.9	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	42.2	43.0	48.8	53.1	58.9	67.0	70.9	71.4	65.5	58.8	48.6	43.0	
	DBStd	12.20	5.79	6.01	6.34	6.00	6.47	6.89	6.02	6.20	5.75	6.27	6.03	5.72	
	HDD50	921	250	203	97	36	7	0	0	0	0	8	96	224	
	HDD65	3924	708	616	502	359	209	58	16	12	62	207	493	682	
	CDD50	3114	7	7	60	128	282	511	647	665	466	281	53	7	
	CDD65	642	0	0	0	1	19	119	198	212	78	15	0	0	
	CDH74	6627	0	0	2	47	299	1356	1944	2110	756	113	0	0	
	CDH80	2891	0	0	0	3	81	592	914	1042	247	12	0	0	
Wind	WSAvg	7.3	6.1	7.5	7.8	7.8	7.6	8.0	8.8	8.0	6.9	6.7	6.7	5.7	
Precipitation	PrecAvg	41.9	4.3	3.9	3.8	4.3	3.4	2.6	1.7	1.6	2.7	3.9	5.4	4.3	
	PrecMax	61.5	14.0	10.6	9.0	8.0	7.9	5.4	4.7	4.5	6.3	12.2	9.4	8.5	
	PrecMin	30.7	0.2	0.8	0.6	1.1	1.2	0.2	0.0	0.2	1.1	0.6	1.4	0.2	
	PrecStd	6.5	3.0	2.5	2.4	1.9	1.5	1.3	1.3	1.2	1.5	2.5	2.1	2.1	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	64.3	72.0	78.9	87.7	97.2	97.2	100.0	92.7	81.6	67.7	59.3
MCWB			52.3	49.3	54.6	59.5	64.3	68.4	67.7	68.5	65.8	61.9	58.1	51.6	
2%		DB	56.5	59.5	68.1	74.0	81.0	91.5	93.6	95.1	86.3	77.1	63.6	56.7	
		MCWB	50.4	49.8	53.9	58.3	62.8	67.0	67.4	67.5	65.2	61.6	55.3	50.0	
5%		DB	53.7	55.8	64.4	69.5	76.7	87.4	89.8	91.1	82.4	73.2	60.6	53.8	
		MCWB	48.6	47.7	52.5	55.9	60.7	65.9	67.4	67.3	64.2	60.3	53.8	48.4	
10%		DB	51.4	53.0	60.6	65.1	72.0	82.5	85.0	85.9	78.4	69.8	57.6	51.6	
		MCWB	47.2	46.4	50.6	54.0	59.2	65.0	67.0	66.8	63.0	59.1	52.3	47.1	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	53.3	53.5	57.5	61.8	67.0	71.3	72.6	72.9	69.2	65.5	59.9	52.9
			MCDB	57.7	60.8	68.7	75.5	82.3	86.6	87.6	91.3	84.6	74.9	65.3	57.4
	2%	WB	51.3	50.9	55.2	59.1	64.0	69.1	70.5	70.7	67.2	63.4	56.5	50.8	
		MCDB	55.3	57.8	65.2	72.1	77.7	85.4	85.3	86.8	81.2	73.4	61.5	55.2	
	5%	WB	49.6	48.9	53.2	56.9	61.9	67.5	69.1	68.8	65.6	61.8	54.7	49.2	
		MCDB	53.3	54.3	61.7	66.8	73.8	83.2	83.6	84.3	78.6	70.6	59.3	53.0	
	10%	WB	47.6	47.2	51.3	54.7	60.0	65.9	67.6	67.3	64.1	60.2	52.9	47.7	
		MCDB	50.7	52.1	59.2	63.5	70.3	79.9	81.6	82.2	75.2	68.0	57.2	50.9	
	Mean Daily Temperature Range	5% DB	MDBR	12.5	15.3	18.2	19.0	20.8	23.0	23.6	24.1	21.7	18.3	13.2	13.0
			MCDBR	16.6	22.5	26.6	28.7	30.5	32.2	33.0	33.7	29.5	24.8	17.9	15.3
MCWBR			11.3	14.0	14.4	14.7	14.0	12.5	11.7	12.0	11.9	12.2	11.0	10.3	
5% WB		MCDBR	14.3	19.9	22.5	25.4	27.1	29.3	28.6	29.8	25.8	20.4	15.6	13.7	
		MCWBR	10.5	13.3	13.0	13.5	13.1	12.3	11.2	11.7	11.1	10.5	10.2	9.9	
Clear-Sky Solar Irradiance	taub	0.303	0.318	0.354	0.377	0.387	0.397	0.387	0.389	0.368	0.345	0.317	0.300		
	taud	2.509	2.458	2.371	2.316	2.320	2.322	2.349	2.350	2.403	2.471	2.508	2.517		
	Ebn at Noon	268	281	281	282	281	278	279	275	273	266	259	258		
	Edh at Noon	23	28	34	39	40	40	38	37	33	27	23	21		
All-Sky Solar Radiation	RadAvg	510	794	1181	1518	1822	2070	2115	1849	1449	949	584	464		
	RadStd	56	93	140	159	134	109	98	94	66	79	61	50		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page